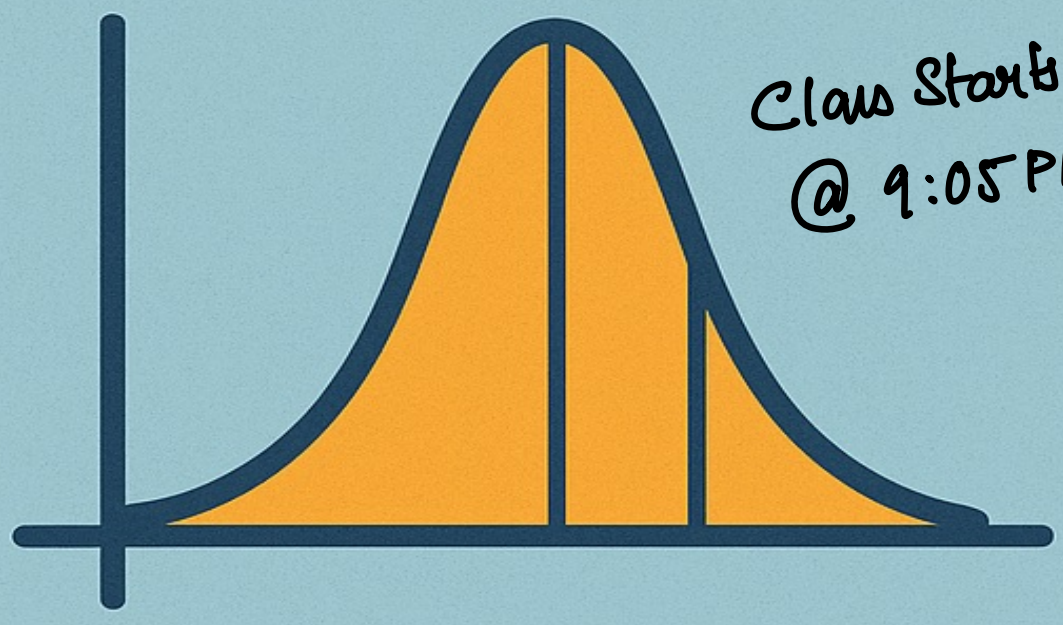


PROBABILITY AND STATISTICS





Lecture 7: Probability Distribution 2

#Agenda

1. Bernoulli Trials ✓
2. Population Vs Sample ✓
3. Sample Statistics ✓
   - → Sample Mean
   - → Sample Variance
   - → Sample Std Deviation
4. Point Estimate ✓
   Standard Error ✓
5. Sampling Techniques ✓
   - → Random Sampling
6. Uniform Distribution
   - ↳ we will start in the next session

$P[X = 2]$ → PMF

$P[60 < X < 70]$ → PDF

$P[X > 30]$ → CDF

Recap

1) PMF → Discrete Random Variable → $P[X=2] = ?$

2) PDF → Continuous Random Variable → $P[60 < X < 70] = ?$

3) CDF → Both (Discreate & Continous)

Binomial Distribution (Discrete Dist)

$P(X \leq 4) = ?$

$P(X \geq 20) = ?$

↳ Event → Only Two possibilities

T/F, Coin Toss,

$$P(X=k) = {}^{n}C_{k} \times P(s)^{k} \cdot P(F)^{n-k}$$

slots — No of trials — 1 − P(s)

Binomial Distribution (Question)

A factory produces LED bulbs, and each bulb has a 5% chance of being defective. A quality inspector randomly selects 20 bulbs from the production line.

What is the probability that exactly 2 bulbs are defective?

$P[X=2] \Rightarrow {}^{n}C_{k} \times P(s)^{n} \times (1-P(s))^{n-k}$

$n = 20$

$P(s) = 5/100$

$k = 2$

$P(F) = 95/100$

$= {}^{20}C_{2} \times (5/100)^{20} \times (95/100)^{18}$

using pmf

Binomial dist $\longrightarrow P[X=2]$

pmf

A multiple-choice quiz has 10 questions, and each question has 4 choices, only one of which is correct. A student guesses the answer to every question.

What is the probability that the student gets exactly 3 answers correct?

$P[X = 3] = \text{binom.pmf}(k=3, n=10, p=1/4)$

A — T
B — F
C — F
D — F

Follow up

$P[X \leq 3] \Rightarrow P[X=1] + P[X=2] + P[X=3]$

cdf

cdf

$\text{binom.cdf}(k=3, n=10, p=1/4)$

$\begin{cases} P(S) = 1/4 \\ P(F) = 1 - P(S) \\ \quad = 3/4 \end{cases}$

$\begin{cases} n = 10 \\ k = 3 \end{cases}$

Bernoulli Trials

$$\underbrace{BD = {}^{n}C_{K} \times P(S)^{n} \times P(F)^{n-K}}$$

When $n=1$

↳ Each trial in a BD, is called Bernoulli trials

2 outcomes in only 1 trial

$$PMF = p^{x} \times (1-p)^{1-x}$$

$x \in 0,1$

$$= p \times 1 \qquad x=1 \quad p$$

$$x=0 \quad = (1-p)$$

BD

$[T, F, T, F \ldots F]$

$n = 10$

Bernoulli Trial

$n = 1$ ✓

$[T]$

Population

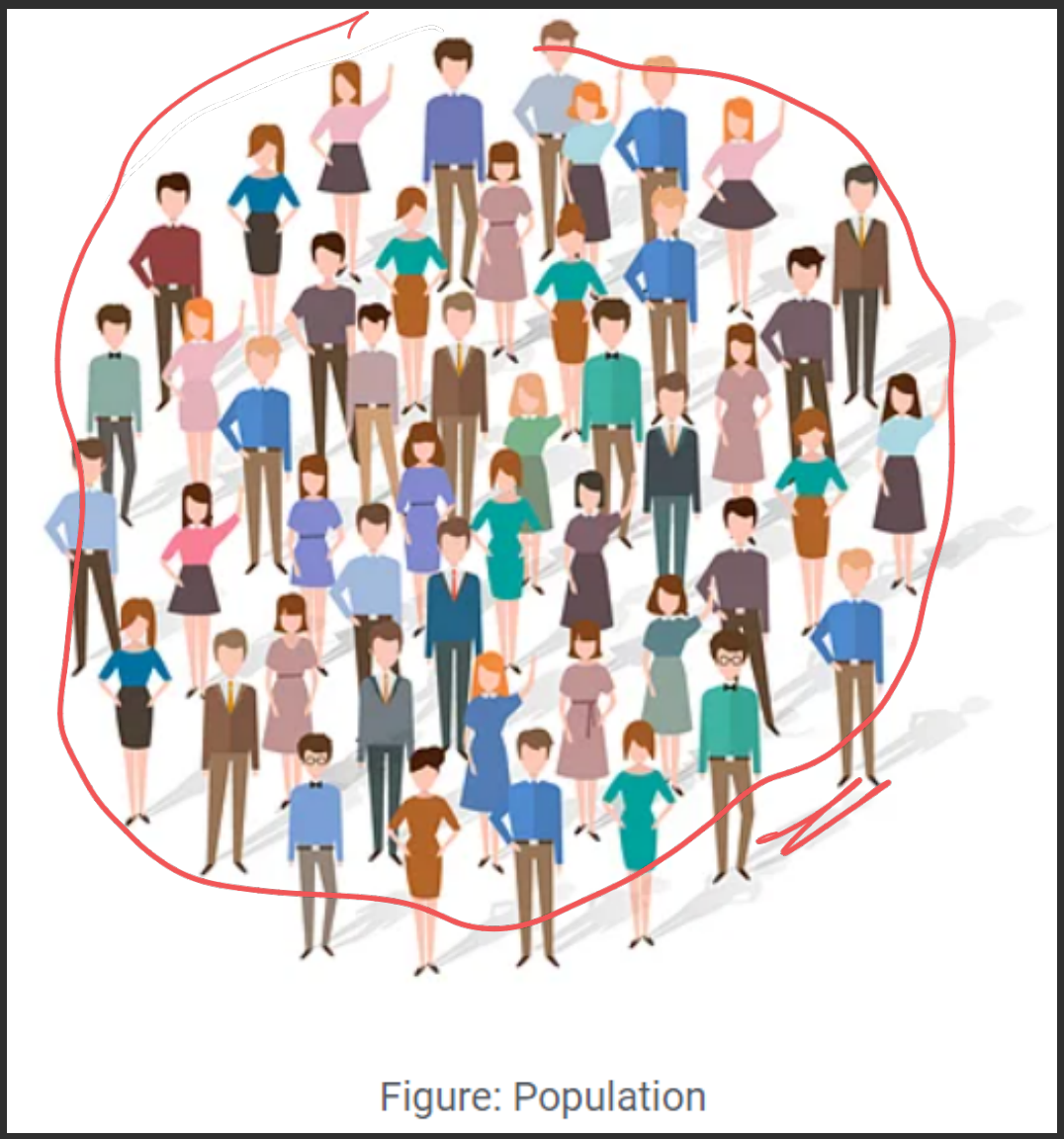


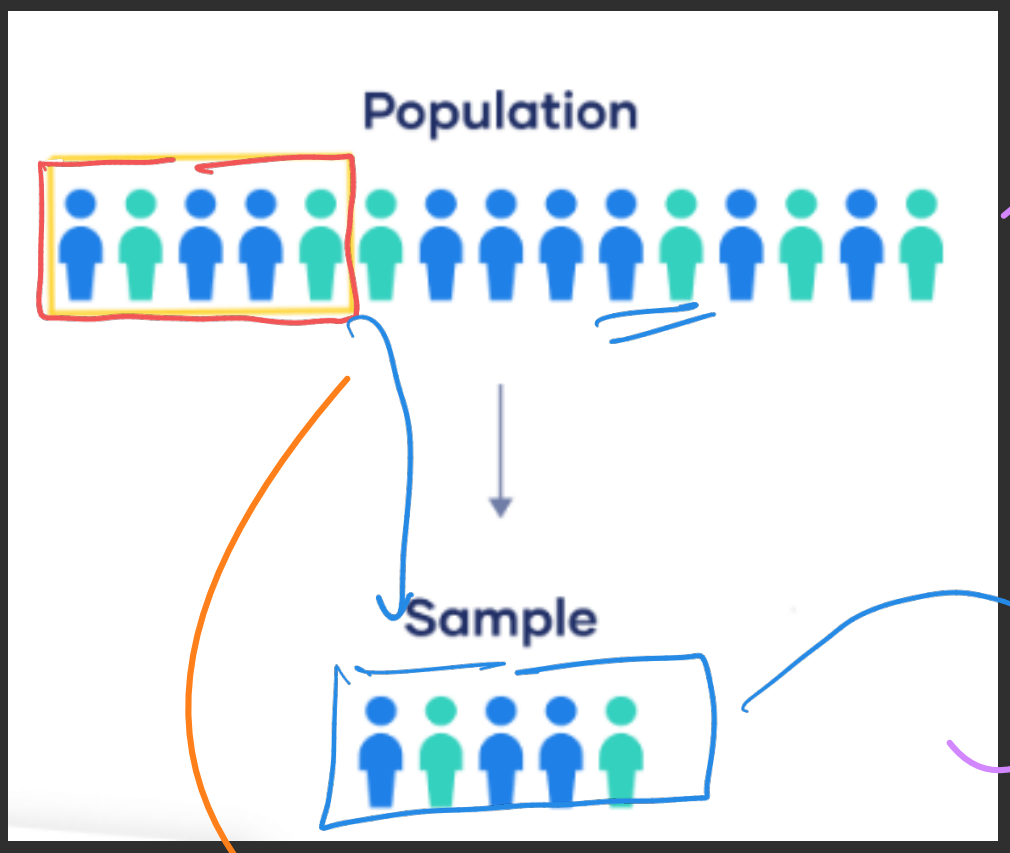
Figure: Population

population

↳ Items under the scope of study

"Height, Weight....."

Sample



Subset of population
smaller group of obs generated from population

{100,000}

subset

{5000}

(smaller group of obs)

Sampling

Process of picking data from population that is called sampling

#Population Vs Sampling



Entire Dataset

Subset of data

(Smaller, manageable version of the whole data)

Why should we use samples instead of studying entire populations?

- Practicality:
  - it's often not feasible to study an entire population due to constraints such as time, resources, and cost.
  - For instance, if you're inspecting a shipment of fresh fruit, you can't inspect every single piece; instead, you take a sample to represent the whole.

- Reduced Error:
  - A well-designed sample can provide an accurate estimate of the population characteristics with significantly less time and effort.

Sampling Statistics

Population

1. Population Mean
2. Population Variance
3. Population SD

Sample

1. Sample mean
2. Sample variance
3. Sample SD

Sampling Statics



Sample Mean

$a = [101, 401, 501, 801, 201 \cdots]$ $n = 10{,}000$

↳ pop^n data

$\mu = \frac{\sum_{i=1}^{n} x_i}{n}$

pop^n mean

dpt from pop^n data

$s = [401, 801, 201 \cdots]$ $n = 1000$

x_i

$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$

↳ sample mean

↳ sample size

Sample Variance

$$\text{pop}^n \text{ var } (\sigma^2) = \frac{\sum_{i=1}^{n} (x_i - \mu)^2}{n}$$

- μ → pop^n mean
- n → pop^n dt
- n → pop^n total data pt

$n = 10K$

$$\text{Samp Var } (s^2) = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$$

- $\bar{x}$ → sample mean
- $n-1$ → sample dt
- $n-1$ → sample total data pt

$n = 1K$

Why n-1 ?

Its called "Bessel's Correction". It important for ensuring that the sample variance provides an unbiased estimate of population variance.

① Degree of freedom: $\text{Var} = (x_i - \bar{x})^2$

↳ sample mean (which is build on top of n samples)

Now, we have n-1 data point can vary freely; the last value is constrained by the othes, as the mean must be same.

$$\frac{1+2+3+4+x}{5} = 32.1 \quad -(1)$$

$x = x$

5 numbers value

$5-1 = \underline{4}$

4 last value

$$\frac{1+2+3+x_1+x_2}{5} = 32.1 \quad X$$

$n-2$

$n-1$

$n-1$

$n-1$

$n-1$ ohs

$5 \to 4$

$$\frac{1+2+3+y+x}{5} = 32.1$$

$$\frac{1+2+3+y+x}{n\ 5} = 32.1$$

Start at
10:12 pm

$$\sum (x_i - \bar{x}) = 0$$

$[4, 7, 9]$

$$\bar{x} = \frac{4+7+9}{3} = 20/3 = 6.66$$

$d = x_i - \bar{x}$

$d_1 = 4 - 6.666 = -2.66$

$d_2 = 7 - 6.666 = 0.333$

$d_1 = 9 - 6.666 = 2.333$

$d_1 + d_2 + d_1 = 0$

$$\sum_{i=1}^{n} (x_i - \bar{x}) = 0$$

5

1, 2, 3, 4, 5

$n \to n$

$n-1 \to n$

$n-1$

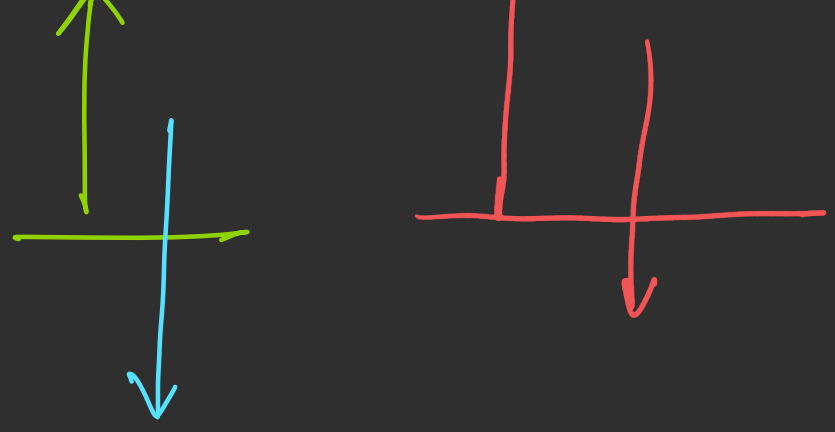
Gender M Female

Red	Green	Blue
1	0	0
0	1	0

$$\text{Samp Var} = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$\bar{x} = \frac{\sum x_i}{n}$

n → sample size



Sample Standard deviation

$$std = \sqrt{var}$$

$$\text{pop std } (\sigma) = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$$

$$\text{sample std } (s) = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

#Quiz 1

There are 45 students in a class.
5 students were randomly selected from this class and their heights (in cm) were recorded as follows:
[131, 150, 140, 142, 152]

Calculate Sample mean and sample variance

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} \Rightarrow \frac{131 + 150 + 140 + 142 + 152}{5} \Rightarrow 143$$

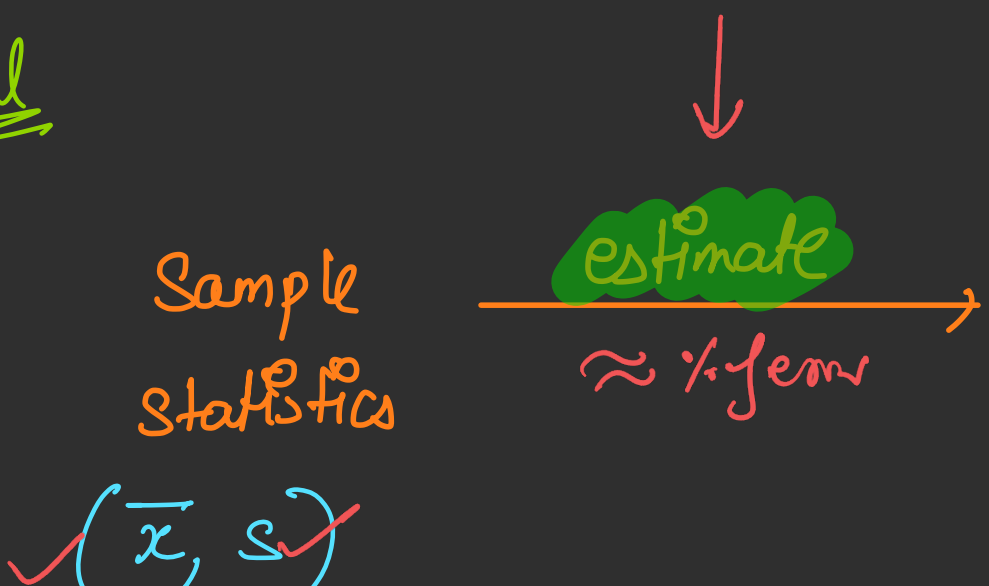
n → sample size

$$\text{sample variance } (s^2) = \frac{\sum_{i=1}^{5} (x_i - \bar{x})^2}{n-1} =$$

$$\rightarrow \frac{(131-143)^2 + (150-143)^2 + (140-143)^2 + (142-143)^2 + (152-143)^2}{5-1}$$

$$\rightarrow \underline{\underline{71}}$$

goal


Sample statistics
$(\bar{x}, s)$
estimate
$\approx$ % error
Population statistic
(μ, σ)
high volume of data

CLT

Point Estimates

Sample statistics $(\bar{x}, s)$ —estimate→ Population statistics (μ, σ)



Florida

specific breed turtle

"weight"

→ 100,000

Assume → 50 Turtle (Sample)

Sample size = 50 (n)

point estimate

sample mean = 150.4 pounds → pop^n mean is also same ?

→ 85 → → 80 - 90 → interval

Let's say, you made a mistake

(1) sample ⟶ mostly heavy turtles (sample mean) ↓ (pop^n mean)

→ ($\bar{x}$ > μ) ↕ any value

(2) sample ⟶ mostly light turtles (sample mean) (pop^n mean)

($\bar{x}$ < μ)

"The way to pick the turtles is imp"

"Sampling is imp"

Sampling is very very important

Sampling Technique

↓

Two Types

DS & ML

↓ (Biased Involve)

Non-probability Based

1. Snowball sampling
2. Judgemental sampling

Probability Based

1. Simple random sampling
2. Systematic sampling
3. Stratified sampling
4. Cluster sampling

→ ML module

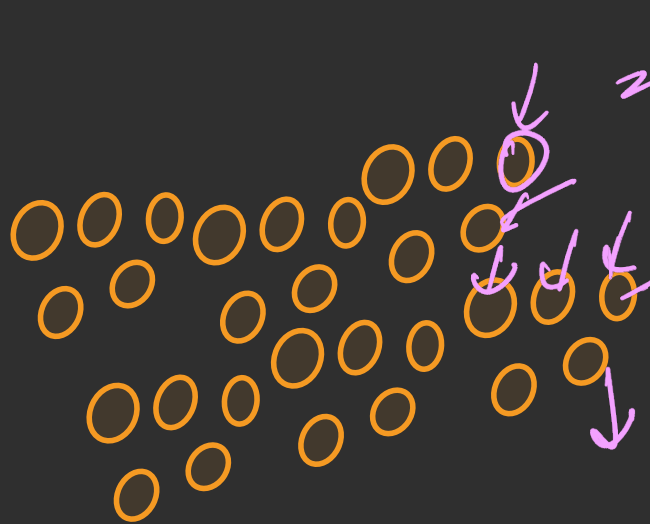
Non-probability sampling techniques are methods in which not all members of the population have an equal chance of being selected for the sample.

Judgmental (Purposive) Sampling: The researcher selects the sample based on their judgment about which subjects are most appropriate for the study. It is often used when specific expertise is needed.

Snowball Sampling: This technique is often used when it’s difficult to identify members of a population. Existing study subjects recruit future subjects from among their acquaintances, leading to a growing sample like a "snowball."

These methods are often used when probability-based sampling is impractical or too costly, though they introduce potential biases.

Simple Random Sampling



P(selecting 1 fruit) = 1/10,000

10,000 fruits

"Simple Random Sampling"

Probability of fruit chosen will remain for all other fruits this is SRS

Sample size = 5

$$\{131, 150, 140, 142, 152\}$$

$$= \frac{131 + 150 + 140 + 142 + 152}{5} = 143$$

$n-1$

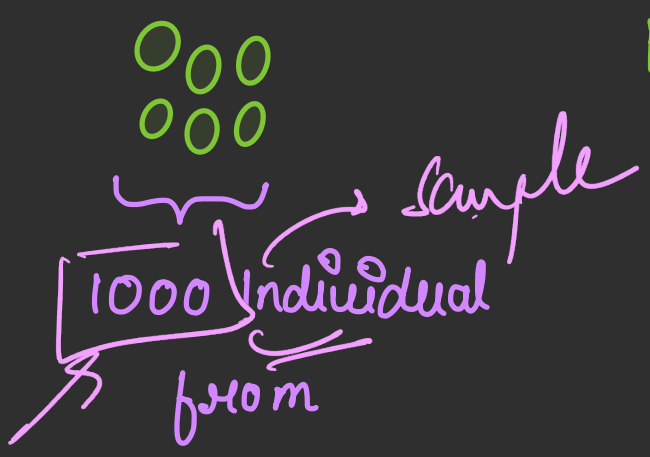
$$\left[(131-143)^2 + (150-143)^2 + (140-143)^2 + (142-143)^2 + (152-143)^2 \right] = 71$$

$$\frac{}{4}$$

For Instance, [Survey data]

Population should always be given

P(Each person getting selected) = 0.01



↓ SRS

Probability of each person getting selected?

Pre-Requisites of this method

1. You have complete list of each Member of the population.
2. You have their contact details
3. You have time + Resources

How To perform "Simple Random Sampling"?

Step1: Population size should be fixed

① 100,000 ✓ ② 120,000 ✗

$(\bar{x} \longrightarrow \mu)$

Sample ⟶ age [18,60] ⟶ 1000

↳ How sure

sample is

VVI

Step2: Determine the Sample size [1000]

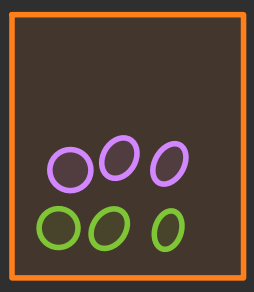
① How sure you are with your result?

↳ 95% certainly → [Confindence Interval]

② You are leaving some chances of error

↳ 5% Moargin of error → precise

Step 3: Randomly select your sample

①  → Pick Randomly → "Lottery Method"

② pop^{n} data → "Random Generator"

↳ label them → [1, 2, 3, 4, ... 10,0000]

"Pick Randomly"

Two Types of Sampling

- Sample with Replacement (Bootstrapping)
  - Used when you have less sample data
- Sample without replacement

$n = 12$

→ 12

sample mean $\xrightarrow{\text{approx}}$ population mean

↳ How to validate?

Standard Error

Q) How to quantify the variability which is caused by the variation in sample statistics across different sample?

→ Standard Error

↳ Deviation in sample mean wrt pop^{n} mean

Formula

sample mean $\xrightarrow{\text{approx}}$ popln mean

$$SE = \frac{\sigma}{\sqrt{n}}$$

Incase popn std is missing use sample std (s)

popln sd sample std

sample size

Deviation in sample mean from the popln mean

SE → 0.01

$\bar{x}$ → 164.2

SE ⇒ 18.64 ↑

$(\bar{x} \longrightarrow \mu)$ $(\bar{x} \neq \mu)$

$\downarrow SE \propto \frac{1}{\sqrt{n}} \uparrow$

$n = 30$

$\sigma = 0.13$

$SE = \sigma / \sqrt{n}$

$= \frac{0.4}{\sqrt{10}}$

$= 0.02$

Sampling statistics

$SE = \frac{\text{population } \sigma}{\sqrt{\text{sample size}}}$

Law of Large Number

$$SE \propto \frac{1}{\sqrt{n}}$$ (Inversely prop to sample size)

$$\downarrow SE \propto \frac{1}{\sqrt{n}} \uparrow$$ (Larger sample size is better)

Quiz 2

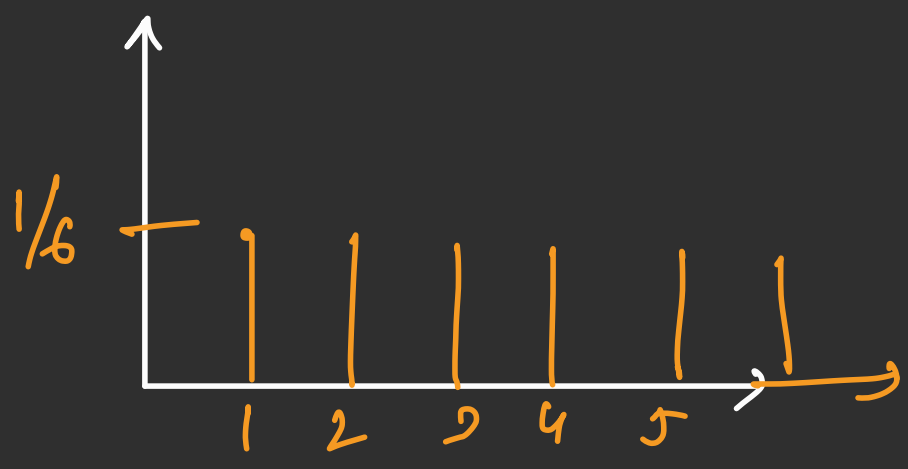
A sample of 30 latest returns on XYZ stock reveals a mean return of $4
with a sample standard deviation of $0.13.
Estimate the SE of the sample mean.

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{s}{\sqrt{n}} = \frac{0.13}{\sqrt{30}} = \underline{\underline{0.02}}$$

Uniform Distribution (Discrete & Continous)

P(every outcome is same)

1. Dice → $P(X=1) = 1/6$, $P(X=2) = 1/6$, ⋮, $P(X=6) = 1/6$



2. Cards : P(Heart) = P(Spade) = P(Diamond) = P(Club)



$0, \underline{0.1}, \underline{0.2}, \underline{0.3}, \underline{0.4}, \underline{0.5}, \underline{0.6} \cdots, \underline{1}$

$P(0.1) = P(0.3) = P(0.6)$

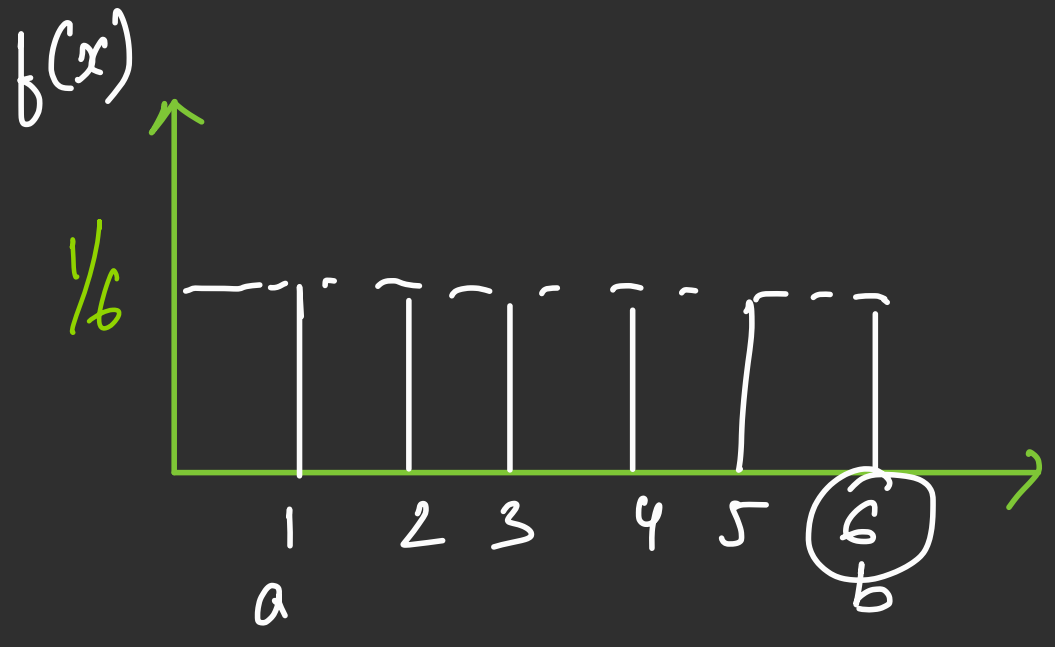
Applicable for both discreate & continuous

① PMF → Discrete uniform distribution

$$PMF(X) = 1/(b-a+1) \quad \text{where} \quad a \le x \le b$$

a → lower limit

b → upper limit



$$P(X=1) = (1/6)$$

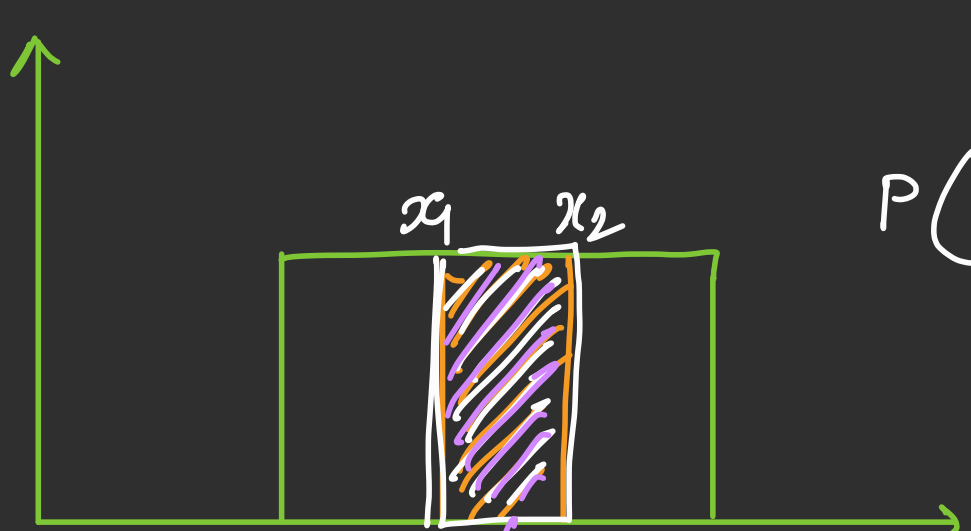
$$P(X=1) = \frac{1}{6-1+1} = \underline{\underline{(1/6)}}$$

Continous Uniform Distribution → PDF

$$PDF(x) = \frac{1}{(b-a)} \quad \text{where } a \le x \le b$$

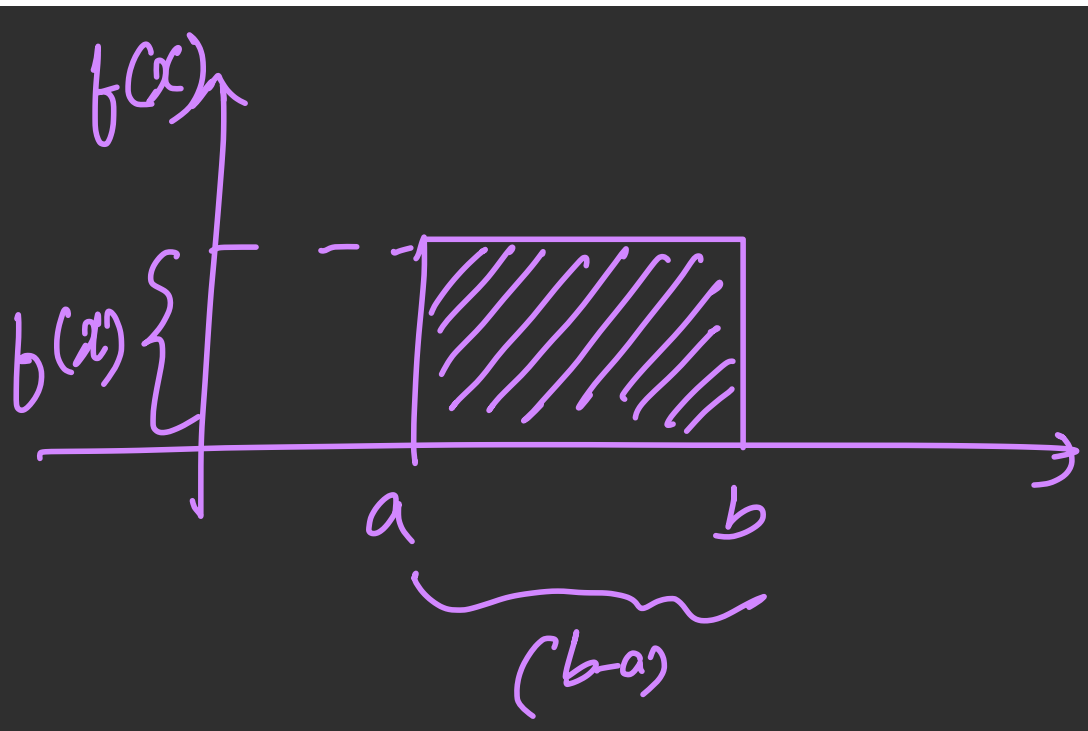
$a \rightarrow$ lower limit

$b \rightarrow$ upper limit



$$P(x_1 \le X \le x_2) = \frac{x_2 - x_1}{(b-a)}$$

Integration $\int_a^b f(x)\,dx$



$$f(x) \times (b-a) = 1$$

$$\begin{cases} f(x) = 1/(b-a) \end{cases}$$

↓

pdf